

AN ELEMENTARY COMPARISON FOR FOUR-ARC REGIONS IN A PLANAR STRIP DENSITY

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ABSTRACT. We consider the planar density equal to one on a horizontal strip and to a constant $\lambda > 1$ outside the strip. We prove that every symmetric four-arc region satisfying the corresponding incidence relation admits a three-arc competitor with the same weighted area and strictly smaller weighted perimeter. The argument applies to every $\lambda > 1$, including the curvature endpoint $h = 1$, and gives an explicit positive lower bound for the perimeter improvement. The proof uses a direct comparison of the quantities $P - hA$ and does not require interval certificates.

1. EVELYN'S INTRODUCTION

NEW STUFF HERE

2. SKILYN'S INTRODUCTION(IN THE WORKS)

The isoperimetric problem, enclose a given amount of space with the least possible boundary, is one of the oldest questions in geometry, but the tools to prove it rigorously are surprisingly recent. De Giorgi's theory of sets of finite perimeter, developed in the 1950s, gave the first fully measure-theoretic proof that the round ball minimizes perimeter for its volume in $\mathbb{R}^n$ [3, 4]. That same machinery turned out to generalize far beyond Euclidean space: replacing ordinary volume and surface area with weighted versions, a positive density f that reweights both, produces the theory of manifolds with density. The best-known example is Gauss space, where the isoperimetric sets are half-spaces [5], a fact that sits at the foundation of concentration-of-measure phenomena in high-dimensional probability. Densities of this kind also turn out to be exactly what Perelman used in his entropy functional for Ricci flow, giving the abstract framework a concrete role in the proof of the Poincaré conjecture. This connection helped motivate a systematic isoperimetric theory for general densities on $\mathbb{R}^n$, culminating in the Log-Convex Density Conjecture of Rosales, Cañete, Bayle and Morgan [6] — that for a smooth, radial, log-convex density, balls about the origin are always isoperimetric — which was fully resolved only in 2015, by Chambers [7]. That resolution is exactly why non-smooth and non-radial densities remain interesting: it says nothing once either hypothesis fails. Piecewise constant densities fail both at once, and are correspondingly much less understood. Cañete, Miranda Jr. and Vittone [1] gave the first systematic study of such densities, including the strip density

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1, \end{cases} \quad \lambda > 1,$$

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which is arguably the simplest density with a genuine jump discontinuity: a minimizer must trade curvature against a change in the cost of boundary length as it crosses a fixed line, a phenomenon the log-convex theory cannot see at all. For a bounded region $E \subset \mathbb{R}^2$ with piecewise smooth boundary, write

$$A_\lambda(E) = \int_E f_\lambda dx dy, \quad P_\lambda(E) = \int_{\partial E} f_\lambda d\mathcal{H}^1$$

for its weighted area and weighted perimeter. Using the Snell-type refraction law satisfied by isoperimetric boundaries at the jump $\{|y| = 1\}$, Cañete–Miranda–Vittone classified the rotationally symmetric candidates for the strip problem into four families: balls contained in the strip, "stadium" regions bounded by two semicircles and two straight segments, three-arc regions, and symmetric four-arc regions. They conjectured that these four families account for every isoperimetric set, according to area:

2.1. Conjecture([1, Conjecture 3.12]). . *There exists $v_0 > \pi$ such that, for the strip density above, the isoperimetric set of area v is a ball for $v \leq \pi$, a stadium region for $\pi \leq v \leq v_0$, and a three-arc region for $v \geq v_0$.* They proved this in full except for one gap: ruling out the symmetric four-arc family as a possible minimizer. This they left as an explicit open problem:

2.1.1. Question ([1, Question 1]). Is it true that a symmetric four-arc region can never be isoperimetric? We answer this question completely. Fix $\lambda > 1$ and let C_h denote the symmetric four-arc region of curvature h , consisting of two circular arcs of radius h^{-1} inside the strip and two congruent circular caps outside it, with exterior half-angle α satisfying $\lambda \cos \alpha = h$; the precise coordinate definition is given in Section 3.

2.1.2. Theorem 1. *For every $\lambda > 1$ and $0 < h \leq 1$, there is a three-arc region E_s , with $0 < s < h$, of the same weighted area as C_h ,*

$$A_\lambda(E_s) = A_\lambda(C_h),$$

whose weighted perimeter is strictly smaller, by an explicit and uniform margin:

$$A_\lambda(C_h) - P_\lambda(E_s) > \frac{\gamma_\lambda}{h} > 0, \quad \gamma_\lambda := \lambda \arccos(1/\lambda) - \sqrt{1 - \lambda^{-2}}.$$

In particular, C_h never minimizes weighted perimeter at fixed weighted area, and Conjecture 3.12 of [1] holds in full. The proof is a direct comparison of the quantities $P_\lambda - hA_\lambda$ for the two families, reducing to the monotonicity of a single auxiliary function. It uses no classification or regularity theorem for arbitrary weighted-perimeter minimizers beyond what [1] already establishes, and it applies uniformly across the entire range $\lambda > 1$, including the curvature endpoint $h = 1$ and the degenerating limit $\lambda \downarrow 1$, where the improvement genuinely vanishes. The accompanying source repository [2] also contains an all-density exclusion for this model. Our purpose here is to give a self-contained, elementary proof of the comparison, with the geometric realization of both families and the endpoint arguments included.

3. INTRODUCTION

Let $\lambda > 1$ and consider the strip density in $\mathbb{R}^2$ given by

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1. \end{cases}$$

For a bounded region $E \subset \mathbb{R}^2$ with piecewise smooth boundary, write

$$\mathcal{A}_\lambda(E) = \int_E f_\lambda dx dy, \quad \mathcal{P}_\lambda(E) = \int_{\partial E} f_\lambda d\mathcal{H}^1.$$

The values of the density on the two interfaces are therefore equal to one. All regions used below have piecewise smooth Jordan boundaries. In particular, their topological boundaries agree with their reduced boundaries up to sets of $\mathcal{H}^1$ -measure zero.

Cañete, Miranda Jr. and Vittone [1] describe four geometric families arising in the strip-density problem and establish that these are the only isoperimetric candidates with vertical reflective symmetry. They are

- (1) balls fully contained in the strip
- (2) pieces of the strip bounded on the left and right sides by semicircles and bounded from the top and bottom by $\{|y| = 1\}$
- (3) sets bounded by three circular arcs with the same radius and a straight line segment (possibly a single point) contained in $\{y = -1\}$ on the bottom. The top arc is contained in $\{y > 1\}$ and the other two are contained in the strip meeting the top arc at an angle determined by Snell's law and meeting the bottom straight line segment tangentially.
- (4) sets bounded by four circular arcs with the same radius with two of them contained within the strip, a top arc in $\{y > 1\}$ and a bottom arc in $\{y < -1\}$, meeting at angles determined by Snell's law.

They conjecture in [1][Conjecture 3.12] that the four-arc candidate cannot be isoperimetric. We consider the symmetric four-arc family, consisting of two circular arcs in the strip and two congruent exterior caps. We denote its member of curvature h by C_h . Thus all four arcs have radius h^{-1} , where $0 < h \leq 1$, and the exterior half-angle satisfies

$$\alpha = \arccos(h/\lambda), \quad \lambda \cos \alpha = h. \quad (3.1)$$

The coordinate definition of C_h is given in Section 4.

Theorem 1. *For every $\lambda > 1$ and $0 < h \leq 1$, there is a three-arc region E_s , with $0 < s < h$, such that*

$$\mathcal{A}_\lambda(E_s) = \mathcal{A}_\lambda(C_h)$$

and

$$\mathcal{A}_\lambda(C_h) - \mathcal{P}_\lambda(E_s) > \frac{\gamma_\lambda}{h} > 0, \quad \gamma_\lambda := \lambda \arccos(1/\lambda) - \sqrt{1 - \lambda^{-2}}. \quad (3.2)$$

Consequently, C_h cannot minimize weighted perimeter at fixed weighted area.

The theorem concerns the explicit symmetric family above. Its proof does not use a classification theorem for arbitrary minimizers. The accompanying source repository [2] also contains an all-density exclusion for this model. Our purpose

here is to give a direct proof of the comparison, with the geometric realization and endpoint arguments included.

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DECLARATION OF AI USE

4. GEOMETRIC FORMULAS

Fix $\lambda > 1$. For $-1 \leq z \leq 1$, define

$$B(z) = \lambda \arccos(z/\lambda) + \arcsin z, \quad D(z) = \sqrt{1 - z^2/\lambda^2} - \sqrt{1 - z^2}. \quad (4.1)$$

For $0 < s \leq 1$, put $q = 2s - 1$ and set

$$A_3(s) = \frac{B(q) + \pi/2 + (2s + 1)D(q)}{s^2}, \quad P_3(s) = \frac{2(B(q) + \pi/2 + D(q))}{s}, \quad (4.2)$$

$$A_4(s) = \frac{2(B(s) + sD(s))}{s^2}, \quad P_4(s) = \frac{4B(s)}{s}. \quad (4.3)$$

These quantities are equivalent to those in [1] through elementary trigonometric identities. The values $A_3(1)$ and $P_3(1)$ will be used only as continuous endpoint values of the formulas. The competitors E_s constructed in the proof always have $s < 1$.

We first verify that these are the weighted areas and perimeters of the relevant regions. For $0 < h \leq 1$, let

$$R = h^{-1}, \quad \alpha = \arccos(h/\lambda), \quad d = D(h)/h.$$

Define the central region and upper cap by

$$K_h = \{(x, y) : -1 \leq y \leq 1, |x| \leq d + \sqrt{R^2 - y^2}\},$$

$$H_h = \{(x, y) : y \geq 1, x^2 + (y - (1 - R \cos \alpha))^2 \leq R^2\}.$$

If $\sigma(x, y) = (x, -y)$, then

$$C_h = K_h \cup H_h \cup \sigma(H_h).$$

At $y = \pm 1$, the half-width of K_h is $\sin \alpha/h$, so the caps and central region meet along their full chords. Interior points of these chords are interior points of the union and contribute no boundary length. The exposed boundary consists of the two strip arcs and two exterior arcs. Hence

$$\mathcal{P}(C_h) = \frac{4 \arcsin h}{h} + \frac{4\lambda\alpha}{h} = P_4(h).$$

The central area and the area of one exterior cap are

$$|K_h| = \frac{4D(h)}{h} + \frac{2\sqrt{1 - h^2}}{h} + \frac{2 \arcsin h}{h^2}, \quad |H_h| = \frac{\alpha - \sin \alpha \cos \alpha}{h^2}.$$

Using (3.1), we obtain

$$\mathcal{A}(C_h) = A_4(h), \quad \mathcal{P}(C_h) = P_4(h). \quad (4.4)$$

These computations remain valid when $h = 1$.

For the three-arc family, let $0 < s < 1$ and set

$$q = 2s - 1, \quad R = s^{-1}, \quad \alpha = \arccos(q/\lambda), \quad b = D(q)/s.$$

Since $D(q) \geq 0$, we have $b \geq 0$. Define

$$K_s^{(3)} = \{(x, y) : -1 \leq y \leq 1, |x| \leq b + \sqrt{R^2 - (y - (R - 1))^2}\},$$

$$H_s^{(3)} = \{(x, y) : y \geq 1, x^2 + (y - (1 - R \cos \alpha))^2 \leq R^2\},$$

and put $E_s = K_s^{(3)} \cup H_s^{(3)}$. The radicand in $K_s^{(3)}$ is nonnegative throughout the strip. Its upper half-width is

$$b + \frac{\sqrt{1 - q^2}}{s} = \frac{\sin \alpha}{s},$$

which agrees with the cap half-chord. Its lower boundary is the exposed segment of length $2b$ on $y = -1$.

The two strip arcs have total length $2(\arcsin q + \pi/2)/s$, and the exterior arc has length $2\alpha/s$. Therefore

$$\mathcal{P}(E_s) = \frac{2\lambda\alpha}{s} + \frac{2(\arcsin q + \pi/2)}{s} + 2b = P_3(s).$$

Direct integration gives

$$|K_s^{(3)}| = \frac{4D(q)}{s} + \frac{\arcsin q + \pi/2 + q\sqrt{1 - q^2}}{s^2}, \quad \lambda|H_s^{(3)}| = \frac{\lambda\alpha - q \sin \alpha}{s^2}.$$

Adding these terms yields

$$\mathcal{A}(E_s) = A_3(s), \quad \mathcal{P}(E_s) = P_3(s). \quad (4.5)$$

When $s < 1/2$, the upper cap has $\alpha > \pi/2$ and is a major circular segment. It is still given by the intersection of the disk with $y \geq 1$, so the construction remains valid. When $s = 1/2$, the lower segment reduces to one point and E_s is the disk of radius two centered at $(0, 1)$. Thus every parameter $0 < s < 1$ gives a valid competitor. All the regions above are bounded, and the finitely many arc junctions have zero $\mathcal{H}^1$ -measure.

5. TWO COMPARISONS AT THE SAME CURVATURE

The proof will use two inequalities: $A_3(h) < A_4(h)$ and a strict comparison of $P_i(h) - hA_i(h)$ for $i = 3, 4$. Both follow from the same auxiliary function.

For $-1 \leq z \leq 1$, set

$$S(z) = B(z) - zD(z),$$

and, for $0 \leq u \leq 1$, define

$$J(u) = 2S(u) - S(2u - 1) - \frac{\pi}{2}. \quad (5.1)$$

Lemma 1. *For $0 < u \leq 1$, one has*

$$J(u) \geq \gamma_\lambda > 0, \quad J(u) - uJ'(u) \geq \gamma_\lambda. \quad (5.2)$$

At $u = 1$, J' denotes the continuous extension of its derivative from $(0, 1)$.

Proof. For $-1 < z < 1$, differentiation gives

$$D'(z) = \frac{z}{\sqrt{1-z^2}} - \frac{z}{\lambda\sqrt{\lambda^2-z^2}}, \quad B'(z) + D(z) = zD'(z).$$

It follows that

$$S'(z) = -2D(z). \quad (5.3)$$

The function D is even, and $D'(z) > 0$ for $0 < z < 1$. Indeed,

$$\lambda\sqrt{\lambda^2-z^2} > \sqrt{1-z^2} \quad (|z| < 1).$$

In particular, $D'(z) < 0$ when $-1 < z < 0$.

The trigonometric identities in (4.1) give

$$S(-z) = \lambda\pi - S(z), \quad S(0) = \frac{\lambda\pi}{2}.$$

Consequently,

$$J(0) = J(1) = S(1) - \frac{\pi}{2} = \lambda \arccos(1/\lambda) - \sqrt{1-\lambda^{-2}} = \gamma_\lambda. \quad (5.4)$$

To see that this number is positive, put $\theta = \arccos(1/\lambda) \in (0, \pi/2)$. Then

$$\gamma_\lambda = \lambda\theta - \sin\theta > 0,$$

since $\sin\theta < \theta < \lambda\theta$.

Using (5.3), we compute

$$J'(u) = 4(D(2u-1) - D(u)), \quad 0 < u < 1. \quad (5.5)$$

Since D is even and strictly increasing on $[0, 1]$, this derivative is positive for $0 < u < 1/3$ and negative for $1/3 < u < 1$. Together with (5.4), this proves the first inequality in (5.2).

Let $H(u) = J(u) - uJ'(u)$. For $1/3 \leq u \leq 1$, the sign of J' gives

$$H(u) \geq J(u) \geq \gamma_\lambda.$$

For $0 < u < 1/3$, we have $2u-1 < 0$, and hence

$$J''(u) = 8D'(2u-1) - 4D'(u) < 0.$$

Therefore $H'(u) = -uJ''(u) > 0$ on this interval. Formula (5.5) extends continuously to $u = 0$ with a finite value, so

$$\lim_{u \downarrow 0} H(u) = J(0) = \gamma_\lambda.$$

This proves the second inequality. All endpoint statements follow by continuity; no derivative of a square root is taken at $u = 1$. $\square$

Proposition 1. *For every $\lambda > 1$ and $0 < h \leq 1$,*

$$A_4(h) - A_3(h) \geq \frac{\gamma_\lambda}{h^2} > 0, \quad (5.6)$$

$$(P_4(h) - hA_4(h)) - (P_3(h) - hA_3(h)) \geq \frac{\gamma_\lambda}{h} > 0. \quad (5.7)$$

Proof. Let $q = 2h - 1$. Substituting (4.2)–(4.3) gives

$$\begin{aligned} h(P_4(h) - hA_4(h)) &= 2S(h), \\ h(P_3(h) - hA_3(h)) &= S(q) + \pi/2. \end{aligned}$$

Thus the difference of these two expressions is $J(h)$. A second direct calculation, using (5.5), yields

$$\begin{aligned} h^2(A_4(h) - A_3(h)) &= 2B(h) - B(q) - \pi/2 + 2hD(h) - (2h + 1)D(q) \\ &= J(h) - hJ'(h). \end{aligned}$$

The two claims now follow from Lemma 1. $\square$

6. THE EQUAL-AREA COMPARISON

We next record the variational identity needed to pass from the same-curvature inequalities to an equal-area competitor.

Lemma 2. *The functions in (4.2) are continuous on $(0, 1]$ and differentiable on $(0, 1)$. They satisfy*

$$P'_3(s) = sA'_3(s), \quad 0 < s < 1, \quad (6.1)$$

and $A_3(s) \rightarrow +\infty$ as $s \downarrow 0$.

Proof. Continuity and differentiability follow directly from (4.1)–(4.2). To check (6.1), let $q = 2s - 1$ and put

$$N(s) = B(q) + \pi/2 + (2s + 1)D(q), \quad M(s) = 2(B(q) + \pi/2 + D(q)).$$

Then $A_3 = N/s^2$ and $P_3 = M/s$. The identity $B'(q) + D(q) = qD'(q)$ gives

$$N'(s) = 8sD'(q), \quad M'(s) = 8sD'(q) - 4D(q), \quad 2N(s) - M(s) = 4sD(q).$$

It follows that

$$s^2(P'_3(s) - sA'_3(s)) = s(M'(s) - N'(s)) + 2N(s) - M(s) = 0.$$

Finally, the numerator of $A_3(s)$ tends to

$$B(-1) + \pi/2 + D(-1) = \lambda \arccos(-1/\lambda) + \sqrt{1 - \lambda^{-2}} > 0$$

as $s \downarrow 0$, proving the last claim. $\square$

Proof of Theorem 1. Fix $0 < h \leq 1$ and let $V = A_4(h)$. By (5.6),

$$A_3(h) < V.$$

Choose $0 < \varepsilon < h$ sufficiently small that $A_3(\varepsilon) > V$. The intermediate value theorem shows that

$$Z = \{s \in [\varepsilon, h] : A_3(s) = V\}$$

is nonempty. It is compact, so it has a largest element r . The strict inequalities at the endpoints imply

$$0 < \varepsilon < r < h \leq 1.$$

Moreover,

$$A_3(s) < V \quad \text{for } r < s \leq h. \quad (6.2)$$

Indeed, equality would contradict the choice of r . If $A_3(s) > V$ at some such point, continuity and $A_3(h) < V$ would produce an equal-area root to its right, again contradicting maximality.

Define

$$Q(s) = P_3(s) + s(V - A_3(s)), \quad r \leq s \leq h.$$

By Lemma 2,

$$Q'(s) = V - A_3(s) > 0 \quad (r < s < h).$$

Since Q is continuous on $[r, h]$, it follows that

$$P_3(r) = Q(r) < Q(h) = P_3(h) + h(A_4(h) - A_3(h)).$$

Combining this inequality with (5.7) gives

$$P_4(h) - P_3(r) > \frac{\gamma_\lambda}{h} > 0.$$

The geometric identities (4.4) and (4.5) now give the desired competitor E_r . When $h = 1$, the same proof applies because differentiation is used only on $(r, 1)$ and Q is continuous at 1. $\square$

Remark 1. *The proof selects the largest equal-area root below h . It therefore requires neither uniqueness of the root nor global monotonicity of A_3 . The possible change from a minor to a major exterior cap causes no difficulty because the construction of E_s is valid for every $0 < s < 1$.*

Remark 2. *The lower bound γ_λ tends to zero as $\lambda \downarrow 1$, but remains strictly positive for every $\lambda > 1$. At $\lambda = 1$ the two scalar families have $A_i(h) = \pi/h^2$ and $P_i(h) = 2\pi/h$, and strict improvement is not possible. The density hypothesis is therefore essential.*

Remark 3. *The result excludes the explicitly defined four-arc regions under either the complete-boundary perimeter used here or the reduced-boundary perimeter for these same regions. Applying it to an arbitrary isoperimetric set requires a separate argument identifying that set, up to the relevant equivalence, with the geometric family under consideration. No such classification is asserted in this note.*

REFERENCES

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